

Topic 2.3 Knowledge Representation and Reasoning using First Order [Predicate] Logic | Calculus (FOPL, FOL, FOPC)

a) FOL

- Founded on PL
- More expressive than PL
- Well studied, formalized and mathematically analyzed
- Subsumes or makes foundation of many other knowledge representation languages

b) FOL Syntax: [Rules describing structural units]

page 244 (7.4.1 Syntax)

1-5. Sentence $\rightarrow$ AtomicSentence $|$ $\neg$ Sentence $|$ (Sentence) $|$
Sentence Connective Sentence $|$
Quatifier Variable, Variable, ... Sentence

6-7. AtomicSentence $\rightarrow$ Predicate (Term, Term, ...) $|$ Term = Term

8-10. Term $\rightarrow$ Constant $|$ Variable $|$ Function(Term, Term, ...)

11-14. Connective $\rightarrow \wedge | \vee | \Rightarrow | \Leftrightarrow$

15-16. Quantifier $\rightarrow \forall | \exists$ **17** Variable $\rightarrow x | y | z | \dots$

18 Predicate $\rightarrow$ Father $|$ Subset $|$ Greater $| \dots$

19 Constant $\rightarrow$ Karim $|$ AUST $|$ X1 $| \dots$

20 Function $\rightarrow$ Sum $|$ Log $|$ MotherOf $| \dots$

- **Term** – object, Complex term – complicated object
- **Ground term** – a term without a variable

- **Examples**

- MotherOf (Rina) = Bina
 - syntactically correct FOL sentence

- $\forall x, y, z ((\text{Father}(x, y) \wedge \text{Father}(y, z)) \Rightarrow \text{Grandfather}(x, z))$
 - syntactically correct FOL sentence

**** Analyze the two sentences above in ‘Top-Down’ and ‘Bottom-Up’ fashions.**

c) Semantics of FOL

Rules :

- to **interpret** expressions (sentences), that is,
- to find **truth values** or **matches** by fixing **substitutions** or **binding lists** (lists of 'variable / term' pairs).
- Example from **Kinship Domain**:

TELL (KB, { Father(Karim, Rahim),
 Father(Rahim, Selim),
 Father(Rahim, Halim),
 $\forall x, y, z ((\text{Father}(x, y) \wedge \text{Father}(y, z)) \Rightarrow \text{Grandfather}(x, z))$
 })

1. ASK (KB, Father(Rahim, Selim))
[Is Rahim the father of Selim?]
Returns: **True**

2. ASK (KB, $\exists x, y$ Grandfather(x, y))
[Who are grandfather of whom?]
Returns:
{x / Karim, y / Selim, y / Halim} - **substitution**